

14.2 $A = 12 \text{ cm} = 0.12 \text{ m}$
 $T = 2.0 \text{ s}$
 $\omega = \frac{2\pi}{T} = \pi \text{ rad/s}$
 $t = 0.4 \text{ s}, x_0 = 0.06 \text{ m}, v_0 > 0$

(1) $x_0 = A \cos \varphi \Rightarrow \cos \varphi = \frac{1}{2}$
 $v_0 = -A\omega \sin \varphi \Rightarrow \sin \varphi = -\frac{\sqrt{3}}{2}$
 $\varphi = -\frac{\pi}{3}$

(2) $x = 12 \cos(\pi t - \frac{\pi}{3}) \text{ (cm)}$

At $t = 0.5 \text{ s}$

$x = 12 \cos(\frac{\pi}{2} - \frac{\pi}{3}) = 6\sqrt{3} \text{ cm}$

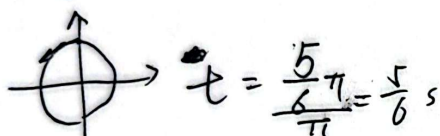
$v = -12\pi \sin(\frac{\pi}{2}) = -6\pi \text{ cm/s}$

$a = -\omega^2 x = -6\sqrt{3}\pi^2 \text{ cm/s}^2$

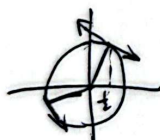
(3) $x = -6.0 \text{ cm}$

$|v| = \omega \sqrt{A^2 - x^2} = 6\sqrt{3}\pi$

$a = -\omega^2 x = 6\pi^2 \text{ cm/s}^2$



14.4

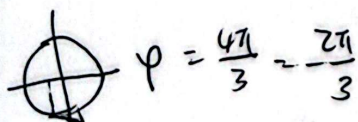


$\Delta \varphi = \frac{2\pi}{3}$

14.5

$A = 2 \text{ cm}$

$v_m = A\omega \Rightarrow \omega = \frac{v_m}{A} = 5 \text{ rad/s}$



$x = 2 \cos(5t - \frac{2\pi}{3}) \text{ (cm)}$

14.6

(1) $\omega = \sqrt{\frac{g}{L}} = \sqrt{\frac{9.8}{1}} \approx 3.13 \text{ rad/s}$

$v_0 = \frac{1}{m} \approx 0.25 \text{ m/s}$

$x_0 = 0$

$A = \frac{v_0}{\omega} = \frac{0.25}{3.13} = 8.0 \text{ cm}$

$\varphi = -\frac{\pi}{2}$

(2) $v_0 = -0.25 \text{ m/s}$

$\varphi = \frac{\pi}{2}$

14.7

$T = 2\pi \sqrt{\frac{M}{k}}$

$M' = m_0 + m$

$T' = 2\pi \sqrt{\frac{m_0 + m}{k}}$

(1) $x = A$ 处速度为 0.

$E_{\text{弹}} = \frac{1}{2} k A^2$

$A' = A$

(2) $v_0 = \omega_0 A = \sqrt{\frac{k}{m_0}} A$

$m_0 v_0 = (m_0 + m) v'$

$v' = \frac{m_0}{m_0 + m} v_0$

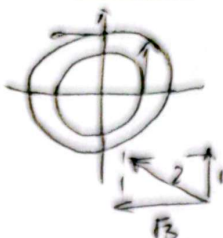
$\frac{1}{2} (m_0 + m) (v')^2 = \frac{1}{2} k (A'_2)^2$

$(A'_2)^2 = \frac{m_0}{m_0 + m} A^2$

$A'_2 = A \sqrt{\frac{m_0}{m_0 + m}}$



14.20



$$A = \sqrt{A_1^2 + A_2^2} = 2$$

$$\tan \varphi_0 = \frac{A_1 \sin \varphi_1 + A_2 \sin \varphi_2}{A_1 \cos \varphi_1 + A_2 \cos \varphi_2} = \frac{1}{\sqrt{3}}$$

$$\varphi_0 = \frac{\pi}{3}$$

$$x = 2 \cos(\omega t + \frac{\pi}{3})$$

4.21.



$$\begin{aligned} A^2 &= A_1^2 + A_2^2 + 2A_1A_2 \cos \frac{2\pi}{3} \\ &= 25 \times 10^{-9} + 24 \times 10^{-9} \cdot \frac{1}{2} \\ &= 37 \times 10^{-9} \text{ m}^2 \end{aligned}$$

$$A = \sqrt{37} \times 10^{-2} \text{ m} \approx 0.0608 \text{ m}$$

$$\tan \varphi_0 = \frac{A_1 \sin \varphi_1 + A_2 \sin \varphi_2}{A_1 \cos \varphi_1 + A_2 \cos \varphi_2} = \frac{1}{7\sqrt{3}}$$

$$\varphi_0 = \arctan\left(\frac{1}{7\sqrt{3}}\right)$$

$$x = 0.061 \cos\left[2t + \arctan\left(\frac{1}{7\sqrt{3}}\right)\right]$$

